DO NOW — RECALL

Paper 2 extended response challenge

Name: _____ Class: _____ Date: _____

1. What is a precondition?

2. What is a loop invariant?

3. What does O(1) auxiliary space mean?

4. For [1,1,2,3,3], count globally distinct values and consecutive runs.

DO NOW — CHECK / OBJECTIVES

Correct and improve

Check page 1 in GREEN PEN. Write corrections in full.

1. Model answer

Requirement that must be satisfied for correctness.

2. Model answer

Property maintained as iterations progress.

3. Model answer

Additional memory remains constant with input size.

4. Model answer

Three distinct values; three consecutive runs.

By the end, you can…

• State the sorted-input precondition and any edge cases.

• Trace changing values with a loop invariant and correct bounds.

• Explain why the selected algorithm is efficient and when another is preferable.

Vocabulary — know these terms

precondition — Required assumption about input.

invariant — Property maintained by a loop.

auxiliary space — Memory used beyond input.

hash set — Structure for expected efficient membership tests.

NEW LEARNING

Read, annotate and explain

Step 1

State the sorted-input precondition and any edge cases.

Step 2

Trace changing values with a loop invariant and correct bounds.

Step 3

Explain why the selected algorithm is efficient and when another is preferable.

Explain one step in your own words, using the correct technical terms.

New Learning · original lesson diagrams

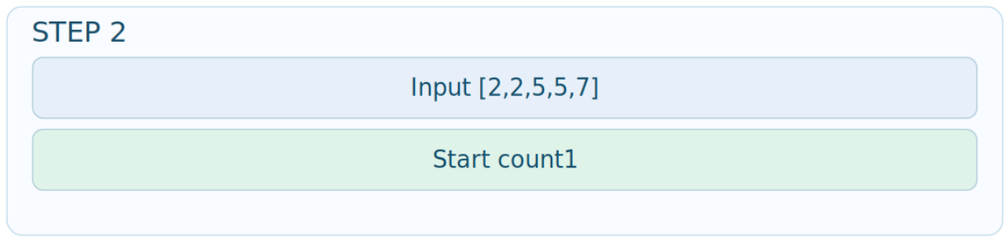
Diagrams appear after the related explanation. Annotate the original source visual.

New Learning



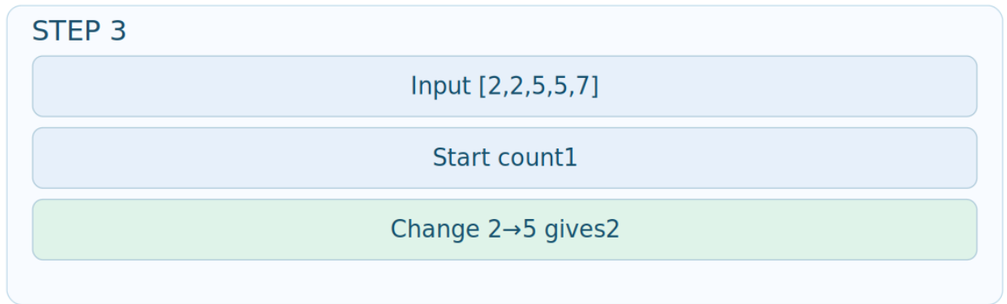
STEP 1

New Learning



STEP 2

New Learning



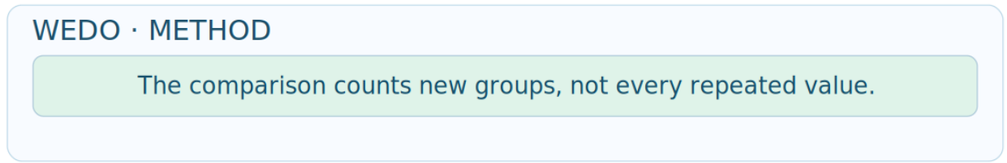
STEP 3

We Do source CHECK · original lesson diagrams

Diagrams appear after the related explanation. Annotate the original source visual.

Trace UniqueCount for [4,4,4,9,10,10].

The comparison counts new groups, not every repeated value.



WEDO · METHOD

WE DO — CHECK AND IMPROVE

Source-grounded model answers

CHECK only after completing the guided practice and Exit Ticket. Use GREEN PEN to improve your method and explain any error.

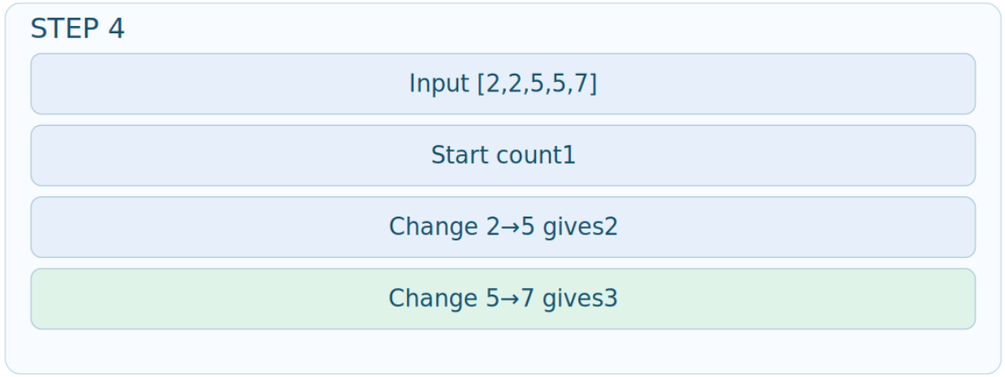
Guided 1 — CHECK

Explain the worked method
The comparison counts new groups, not every repeated value.

New Learning · original lesson diagrams

Diagrams appear after the related explanation. Annotate the original source visual.

New Learning



STEP 4

I DO — WORKED EXAMPLE

Follow the worked method

Trace UniqueCount for [1,1,2,3,3].

Explain the worked method

The loop starts at index1 and compares each item with its predecessor.

EXIT TICKET / REFLECTION

Check your understanding

Complete the actual lesson Exit Ticket before checking.

An adjacent comparison algorithm for sorted input works identically on unsorted input.

Circle: Confident / Getting there / Need help

Explain what you can now do independently.

GREEN PEN — Correct and explain

Choose one error. Write the correct method and explain precisely why your first response was wrong.

YOU DO — INDEPENDENT PRACTICE

Questions · Part 2

Answer independently. Show code, calculations or a justified explanation.

Question 3

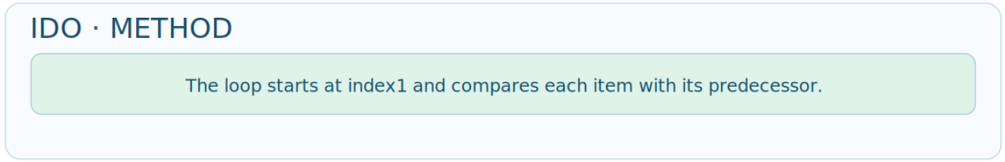
Evaluate adjacent counting versus a hash set when the data is unsorted and contains a million elements. [9]

I Do · original lesson diagrams

Diagrams appear after the related explanation. Annotate the original source visual.

Trace UniqueCount for [1,1,2,3,3].

The loop starts at index1 and compares each item with its predecessor.



IDO · METHOD

WE DO — GUIDED PRACTICE

Try → Discuss → Improve

TRY: Attempt the guided questions in BLACK PEN. DISCUSS: Compare approaches with a partner, explain your reasoning and agree one improvement. IMPROVE: After the Exit Ticket, use the source CHECK model in GREEN PEN to correct and explain errors.

Guided 1 — TRY

Trace UniqueCount for [4,4,4,9,10,10].
Source code / input:
Return3; groups 4,9,10.

YOU DO — INDEPENDENT PRACTICE

Questions · Part 1

Answer independently. Show code, calculations or a justified explanation.

Question 1

Write pseudocode to count distinct values in a sorted array, including empty input. [8]

Question 2

Trace [2,2,5,5,6,9,9] and give time/space complexities. [9]